2025 A2 Computer Systems 245

Question 1:

Answer true or false to the following statements regarding the STM32F407VG Discovery Board. Incorrect answers will earn negative marks. Select "Pass" if you do not want to supply an answer (to earn 0 marks). Choose from True, False or Pass

a) The STM32F407VG features an ARM® 16-bit Cortex®-M4 CPU
b) The board's microcontroller supports floating-point operations in hardware.
c) The board only has **one timer** that can generate PWM signals.
d) The Discovery board includes an **ST-LINK/V2 debugger** integrated on the PCB.
e) The STM32F407VG uses **12-bit registers** for all peripheral control.

Question 2:

Write a C function that will take a byte (unsigned char) as input, and return a byte, according to the following table:

input (as binary):	output:
XXXX 0101	16
XXXX 001X	17
XXXX 0X00	18
XXXX 011X	19

'X' denotes a 'don't care' condition, meaning the value of the particular bit can be either 0 or 1.
Take note: Syntax checking is enabled and you can precheck as many times as you like.
Your code must be compliable to be evaluated. It is not necessary to use any external IDE For example:

Test	Input	Result
4	1110 0010	17
6	0010 0100	18
7	1010 0101	16

Provided code:

```
unsigned char CheckByte(unsigned char input)
                        {
                         ... }
```

Question 3:

The following questions are related to the differences between the BSRR and ODR registers that may be used to change the output logic level of the GPIO pins.

a) If I set a single GPIO output pin to either 0 or 1, which of the following statements is true regarding the number of instructions that are required in the most efficient implementation: (2 marks).
   A. Both BSRR and ODR need same number of instructions
   B. BSRR requires more instructions than ODR
   C. BSRR requires fewer instructions than ODR
   D. Insufficient information

b) You can read from the GPIOA_BSRR register to determine the current output pin logic level. (This question has negative marking) (1 mark)
   True / Waar False / Vals Pass / Slaan oor
c) You can read from the GPIOA_ODR register to determine the output pin logic level. (This question has negative marking) (1 mark)
   True / Waar False / Vals Pass / Slaan oor

Question 4:

The input clock to a timer module is 90 MHz, and the prescaler register TIMx_PSC is set to 999. To generate an interrupt every 0.2 seconds, what should the auto-reload value (TIMx_ARR) be?
179999 17999 18000 180000

Question 5:

The input clock to a timer is 64 MHz, and the prescaler register TIMx_PSC is set to 2047. What is the resulting timer frequency?
31.25 kHz 3125.000 kHz 3.127 kHz 3.000 kHz

Question 6:

a) Which timer mode can be used to generate a single pulse in response to an external event?
   Output Compare
   PWM Mode
   One-Pulse Mode
   Input Capture
b) What kind of signal does a PWM output typically produce?
   Analog sine wave
   Continuous DC signal
   Square wave with adjustable duty cycle
   Triangular wave

Question 7:

Answer true or false to the following statements. Incorrect answers will earn negative marks. Select "Pass" if you do not want to supply an answer (to earn 0 marks).

a) All Serial Communication must be bi-directional.
b) SPI is a full-duplex, point-to-point communication standard
c) UART can be used to communicate between a microcontroller and a user using terminal emulator like Termite.
d) In I2C, each byte must be followed by an Acknowledge bit.
e) The USB Virtual COM port device can be used as USB connected UART to your PC, when using the STM32F407VG Discovery Board

Question 8:

Arrange the serial communication standards from the fastest to slowest
UART SPI I2C

Question 9:

If a UART signal bit-rate is 115200 bits per second, using 8N1 (no parity), what is the data transfer rate in bytes per second?

Question 10:

Given that a UART COM port that has been configured to operate with even parity, one start bit and one stop bit, and transmits the character "t", answer the following questions. Find the ACII table for yourself.

a) What is the 11-bit data frame?
b) How many parity bits are there in a frame?
   0
   1
   2

Question 11:

Choose the correct answers. Incorrect answers will earn negative marks. Select "Pass" if you do not want to supply an answer (to earn 0 marks).

a) SPI supports multiple masters?
   True / Waar False / Vals Pass / Slaan oor
b) SPI only supports 8-bit transfers?
   True / Waar False / Vals Pass / Slaan oor
c) SPI hardware controllers can also be used for transmitting [A] data?
   I²S I²C UART

Question 15:

For an I/O drive strength of 6mA and VDD of 3.3V, calculate the minimum pull-up resistor value in Ohms for a I2C bus signal using the correct values from the table below: It is table 54 from one of the manuals. Round off to 2 significant figures.

Question 16:

For a 0-3.3V ADC, choose the minimum resolution needed for a quantisation error of less than 5 mV.

4-bit 6-bit 8-bit 10-bit 12-bit 14-bit 16-bit 24-bit

Question 17:

True or False: The STM32F407 uses a Flash ADC?

Question 18:

True or false: For SAR ADC, the conversion result is decoded instantly?

Question 19:

For a 0-3.3V 16-bit DAC calculate the analog output value for a digital value of 0xF0E7? (Round your answer to 4 decimal places)

Question 13:

A junior engineer wrote driver functions to initialize and read data from the linear accelerometer in the ST LIS3DSH MEMS sensor via SPI.

You are tasked with fixing the errors in the LIS3DSH_Init and LIS3DSH_ReadXYZ functions.

The current symptom is: It does not work. Correct the code in both the functions so that it initializes and reads the X,Y and Z acceleration from the linear accelerometer in normal mode with an output data rate of 50 Hz. The datasheet is available on stemlearn.sun.ac.za and here : LIS3DSH datasheet.

An excerpt of the shemactic is shown below. You can assume that the GPIO pins and SPI pins have been already configured correctly by the graphical configuration tool.

Take note: This Coderunner question does NOT penalize incorrect checks. Syntax checking is enabled and you can check as many times as you like. It is not necessary to use any external IDE

Code given:

```
void LIS3DSH_Init(void)
{
  HAL_StatusTypeDef res;
  uint8_t TxBuf[10] = {0};
  TxBuf[0] = 0x20 | 0x80;
  TxBuf[1] = 0x13;
  res = HAL_SPI_Transmit(&hspi1, TxBuf, 2, 100);
}

void LIS3DSH_ReadXYZ(int16_t *pData)
{
  HAL_StatusTypeDef res;
  uint8_t TxBuf[10] = {0};
  uint8_t RxBuf[10] = {0};
  TxBuf[0] = 0x18;
  HAL_GPIO_WritePin(GPIOE, GPIO_PIN_3, GPIO_PIN_SET);
  res = HAL_SPI_Transmit(&hspi1, RxBuf, 1, 100);
  res = HAL_SPI_Receive(&hspi1, TxBuf, 3, 100);
  HAL_GPIO_WritePin(GPIOE, GPIO_PIN_3, GPIO_PIN_RESET);
  pData[0] = RxBuf[0];
  pData[1] = RxBuf[2];
  pData[2] = RxBuf[4];
}
```

Question 20:

Consider DMA1, stream 3, with CHSEL = 000. When is the DMA request is generated?

Use the reference manual available here. (Note that this question has negative marking)

SPI2 receives data

UART7 receives data

UART4 transmits data

I2S3 receives data

SPI3 receives data

Question 21:

a) In a design with small discrete movements, use a [A] motor.
   Options: DC / Servo / Stepper
b) When switching relays, include a [B] in parallel with the [C].
   Options: resistor, transistor, capacitor, flyback diode, relay, coil
c) For efficient DC motor speed control, use a [E] to drive a [F].
   Options: DMA, GPIO, PWM signal, H-Bridge, relay

Question 22:

Answer true or false to the following statements . Incorrect answers will earn negative marks.
Select "Pass" if you do not want to supply an answer (to earn 0 marks).

a) Wires or traces carrying differential signals must have equal length, otherwise the one signal will arrive before the other.
   True / Waar False / Vals Pass / Slaan oor
b) Characteristic impedance is the resistance of the wire.
   True / Waar False / Vals Pass / Slaan oor
c) Coaxial cables typically have characteristic impedances of 50 Ω or 75 Ω.
   True / Waar False / Vals Pass / Slaan oor

Question 23:

Select the correct answer in the following statements:

a) In general, [A] flash is used for program code
   NAND NOR
b) [B] flash memory is used for mass storage
   NAND NOR
c) A [C] interface has separate strobe signals for column and row addresses.
   SDRAM NOR NAND
d) [D] information is stored as charge in a capacitor.
   SDRAM NOR NAND

Question 24:

Block diagram picture use some common sense and read instructions carefully

Question 25:

Write an assembly function readUserPin to read the state of PA0 for the STM32F4DISCOVERY development kit as shown in the schematic above.
You can assume the GPIO ports have been configured correctly using the STM32CubeMX and that the initialisation routines have already been called in the main function.
The equivelant C-prototpye would be: uint32_t readUserPin();
Take note:
You may only work with input data register.
This Coderunner question does penalize incorrect checks.

Syntax checking is enabled and you can precheck as many times as you like.
Your code must compile in order to be evaluated. It is not necessary to use any external IDE.
Given code:

```
.syntax unified
/*Use varibales below to control the debug trace.
mem_addr_start=0x40020000
mem_addr_length=32
print_mem_inline=False
mem_display_mode=hex */

readUserPin:
```